MILITARY STANDARD

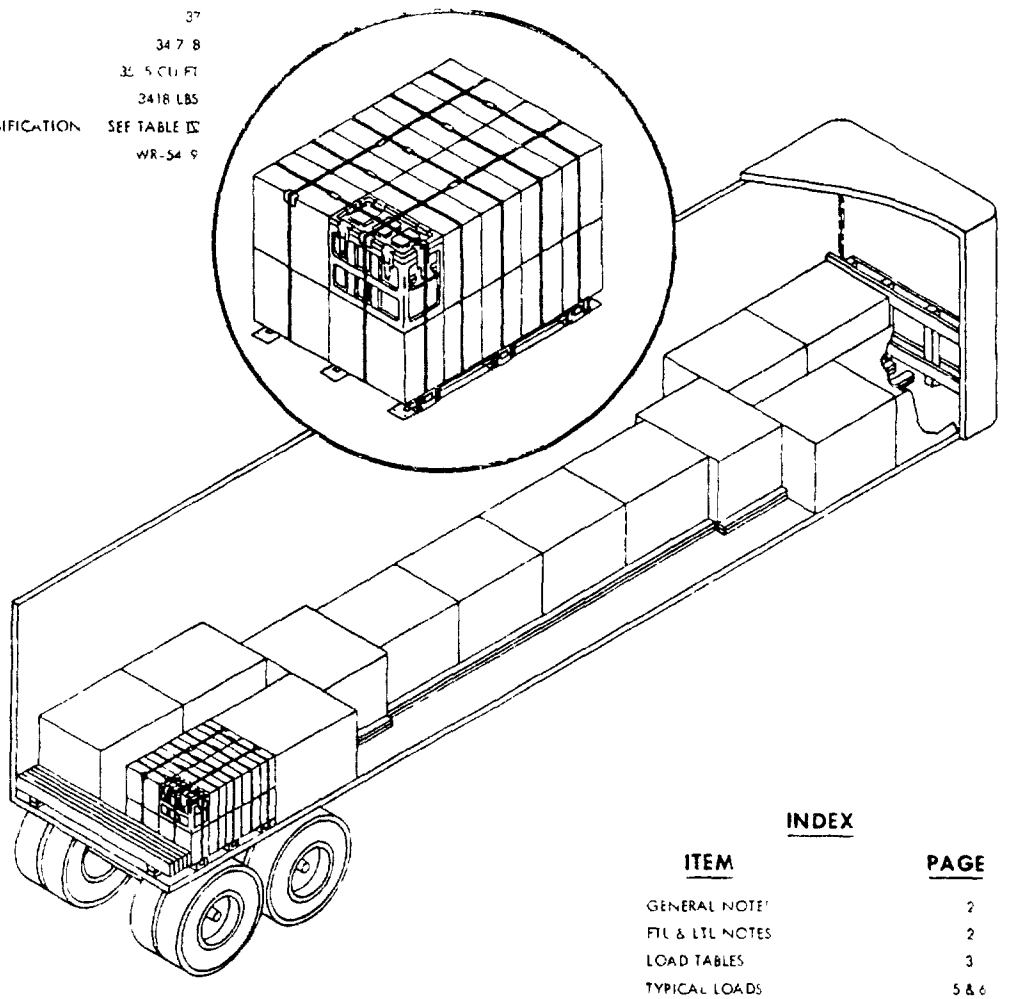
MIL-STD-1320-1/9
(NAVY)

1 MARCH 1977

TRUCKLOADING
CARTRIDGE 20MM, MK 100 SERIES
NOT LINKED, BULK PACK
FOR MK 11 OR MK 12 GUN
FLEET ISSUE UNIT LOAD

UNIT LOAD DATA

LENGTH	47 1/2
WIDTH	37
HEIGHT	34 7/8
CUBE	35.5 CU FT
WT APPROX	3418 LBS
DOT HAZARD CLASSIFICATION	SEE TABLE IX
UNIT LOAD DWC	WR-54.9



INDEX

ITEM	PAGE
GENERAL NOTES	2
FTL & LTL NOTES	2
LOAD TABLES	3
TYPICAL LOADS	5 & 6

NOTES

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING CONTAINERLOADING AND CARLOADING MILITARY STANDARDS REFER TO INDEX TO STANDARDS MIL HDBK 2xx NAVY

FSC 8140

AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENTS ONLY

TECHNICAL DIRECTION AGENT (TDA) — DATE 2/22/77

AIRS/SEA DIRECTION — DATE 2 23 77

ORIGINATOR — DATE 2/23/77

NAVAL WEAPONS HANDLING CENTER
WPNSTA EARLE, NEW JERSEY

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GENERAL NOTES

1. THIS DOCUMENT PROVIDES DETAILED INSTRUCTIONS FOR TRUCKLOADING FLEET ISSUE UNIT LOADS OF THE MK 100 SERIES 20 MM CARTRIDGES (NOT LINKED) IN THE SMALL ARMS AMMUNITION BOX MK 1 MOD 0 THE UNIT LOAD SHALL BE IN ACCORDANCE WITH WR-54/9
2. THE PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40 42, 44 AND 45 FT TRAILERS WITH BOTH NAILABLE AND NON-NAILABLE FLOORS THE TRAILER AXLES SHALL BE LOCATED IN THE WESTERN POSITION (AT THE EXTREME REAR OF THE TRAILER)
3. WHEN THE SPACE AT THE REAR OF THE LOAD IS LESS THAN 12 INCHES USE DETAIL B PAGE 4 FOR THE REAR BLOCKING WHEN THE SPACE IS BETWEEN 12 AND 36 INCHES USE DETAIL D, PAGE 4 DO NOT USE TRAILERS WITH ROLL UP DOORS WHEN BLOCKING BY THIS METHOD
4. THE REAR BLOCKING ASSEMBLIES DETAIL B OR D SHALL FILL THE SPACE BETWEEN THE LADING AND THE TRAILER DOORS WHEN THE DOORS ARE CLOSED
5. WHEN THE TRAILER HAS NON-NAILABLE FLOORS AND/OR SLEEPER PIECES / AND 8 CAN NOT BE USED, SUBSTITUTE SIDE BLOCKING ASSEMBLY DETAIL C THE ASSEMBLY SHALL FILL THE VOID BETWEEN THE LADING AND TRAILER SIDE WALLS A SLIP FIT MAY BE USED PROVIDED THE CUMULATIVE SLACK BETWEEN THE UNIT LOAD AND THE TRAILER WALLS DOES NOT EXCEED ONE INCH
6. WHEN THE TRAILER HAS NON-NAILABLE FLOORS AND/OR SLEEPER PIECE 6 CAN NOT BE USED SUBSTITUTE SWAY BRACE ASSEMBLY DETAIL E INSERT SUPPORTS AGAINST INBOARD SIDE OF VERTICAL SUPPORTS POSITION STRINGERS AGAINST VERTICAL SUPPORTS AND NAIL IN PLACE WITH THREE 10d NAILS
7. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS
8. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS TO INSIDE OF TRAILER IN AN ACCESSIBLE AREA, CLOSE AND SEAL TRAILER DOORS ATTACH APPROPRIATE PLACARD FOR THE TYPE OF 20 MM CARTRIDGE BEING SHIPPED TO BOTH SIDES FRONT AND BACK OF TRAILER (SEE TABLE IV PAGE 4,
9. APPLICABLE MATERIAL SPECIFICATIONS DUNNAGE LUMBER MM-L-751 NAILS FF-N-105 TYPE II STYLE 10 COMMON BRIGHT STRAPPING QQ-S-781 TYPE I CLASS A PLYWOOD COMMERCIAL STANDARD CS 45 GRADE INTERIOR C-D
10. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT MIL STD-1320 (NAVY)

FTL AND LTL 40, 42, 44 AND 45 FT TRAILERS (VAN)

WARNING

THIS STANDARD PROVIDES FOR INCREASED LOADINGS WHICH PRODUCE HEAVIER GROSS TANDEM AXLE WEIGHTS AND GROSS VEHICLE WEIGHTS THESE INCREASED WEIGHTS ARE NOT AUTHORIZED IN ALL 50 STATES BEFORE LOADING MORE THAN 12 UNIT LOADS IN A TRAILER DETERMINE THAT THE LIMITATIONS OF THE NOTES FOR TABLES I II & III ARE MET

1. TABLES I II AND III PROVIDE THE CORRECT LOAD PATTERNS FOR THE LENGTH OF THE TRAILER BEING LOADED AND THE ALLOWABLE GROSS TANDEM AXLE WEIGHT AND GROSS VEHICLE WEIGHTS
2. TO DETERMINE THE CORRECT LOAD PATTERN TO USE SELECT THE CORRECT TABLE FOR THE LENGTH AND WEIGHT OF TRAILER BEING LOADED OPPOSITE THE NUMBER OF UNIT LOADS TO BE LOADED FIND THE CORRECT LOAD PATTERN MAKE CERTAIN THAT THE QUANTITY BEING SHIPPED IS COMPATIBLE WITH THE WEIGHT LAWS OF THE STATES THROUGH WHICH THE SHIPMENT IS ROUTED BLOCK AND BRACE USING THE PRINCIPLES SHOWN ON PAGES 5 & 6

NOTE

LIGHT WEIGHT TRACTOR-TRAILER - TARE WEIGHT OF TRAILER TANDEM AXLES 6 800 LBS MAXIMUM AND THE DRIVE TANDEM AXLES 10,500 LBS MAXIMUM

STANDARD WEIGHT TRACTOR-TRAILER - TARE WEIGHT OF TRAILER TANDEM AXLES 7 800 LBS MAXIMUM AND THE DRIVE TANDEM AXLES 11,500 LBS MAXIMUM

THE ABOVE MAXIMUM WEIGHTS MAY BE ADJUSTED WHEN EXPERIENCE INDICATES THAT THE WEIGHT RESTRICTIONS CAN BE MET

3. THE TRUCKLOAD SHOWN ON PAGE 5 DOES NOT LIMIT SHIPMENT TO THE NUMBER OF UNIT LOADS SHOWN IT IS PRESENTED TO SHOW THE BASIC PRINCIPLES OF BLOCKING ONLY THE LOAD PATTERN TABLES SHOW THE CORRECT NUMBER OF UNIT LOADS FOR THE LENGTH AND WEIGHT OF THE TRAILER BEING LOADED AND THE WEIGHT RESTRICTIONS OF THE ROUTING
4. WHEN SHIPPING LESS THAN THE AMOUNTS SHOWN IN THE LOAD TABLES FOLLOW THE PROCEDURES SHOWN ON PAGE 6 DO NOT USE TRAILERS WITH NON-NAILABLE FLOORS

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TABLE I

0 FT TRAILER

LOAD PATTERN

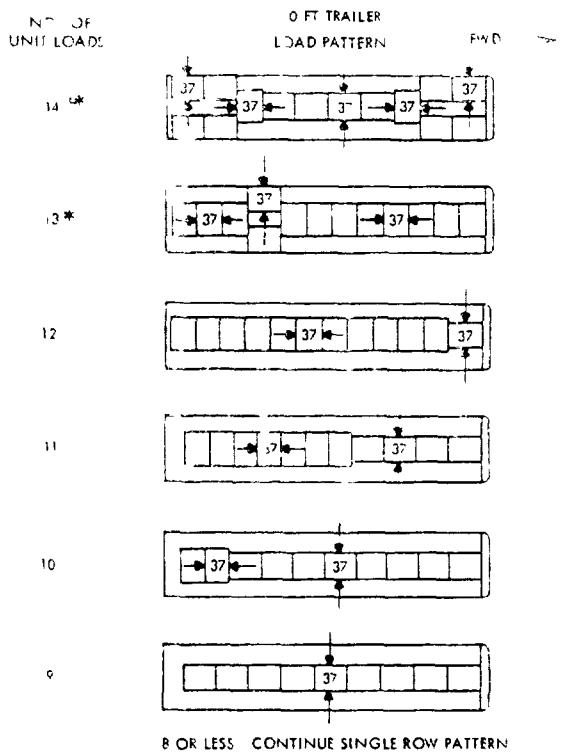


TABLE II

42 FT TRAILER

LOAD PATTERN

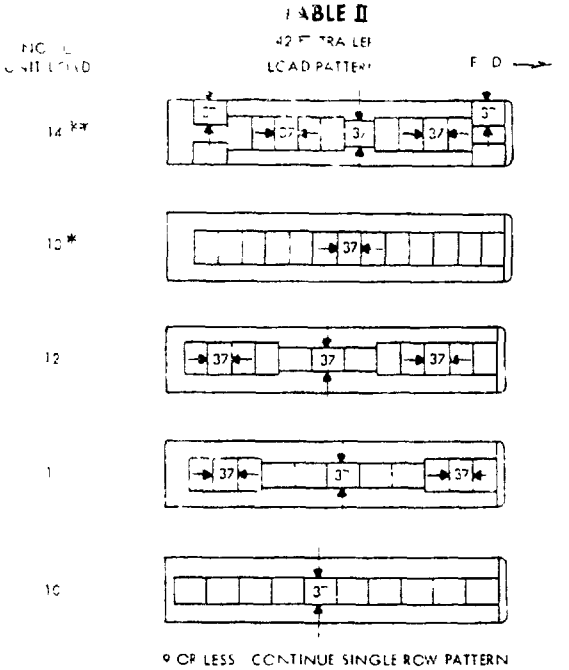
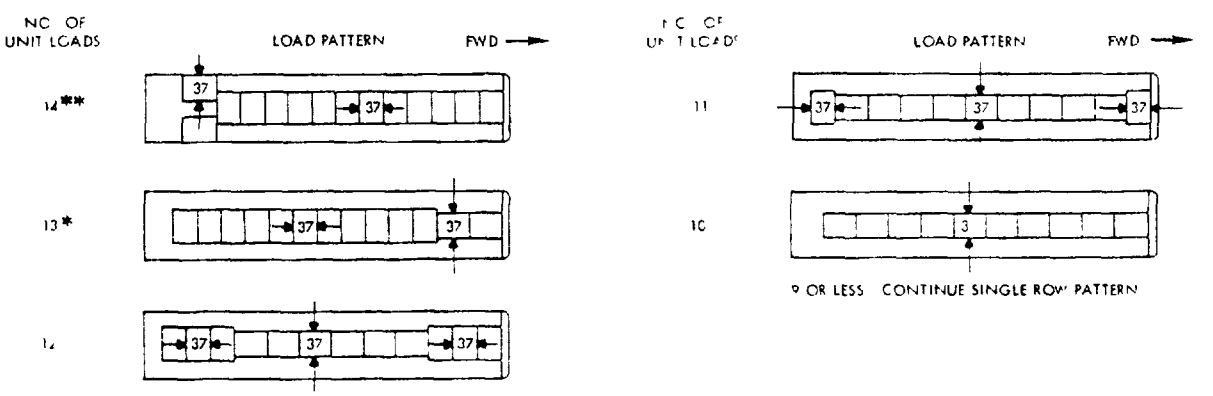


TABLE III

44 & 45 FT TRAILERS

LOAD PATTERN

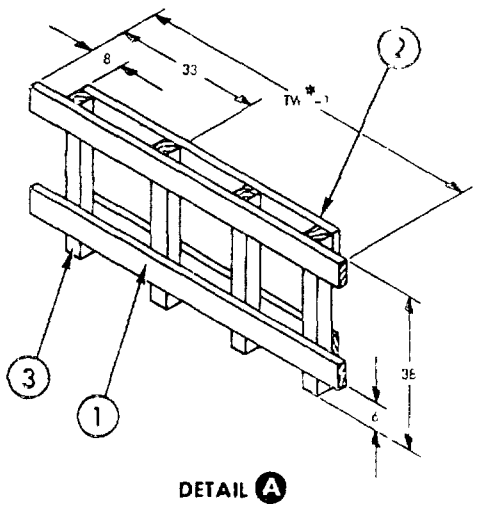


** AUTHORIZED WHEN ROUTING IS THROUGH STATES PERMITTING 34 000 LBS OR MORE ON THE TANDEM AXLES AND A GROSS VEHICLE WEIGHT OF 80 000 LBS ONLY LIGHT WEIGHT TRACTOR-TRAILERS SHALL BE USED

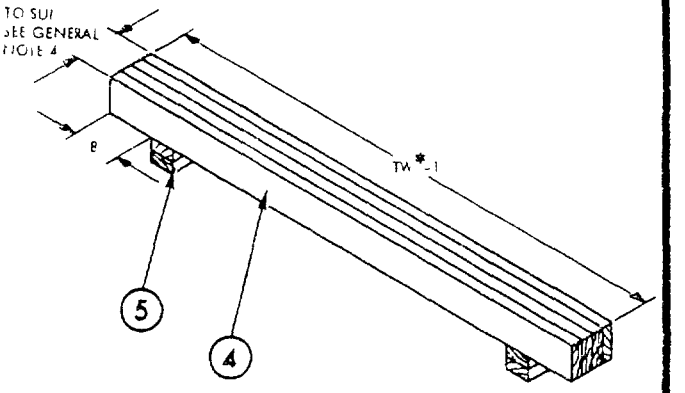
* AUTHORIZED FOR LIGHT WEIGHT TRACTOR-TRAILERS ONLY WHEN ROUTING IS THPOUGH STATES LIMITING TANDEM AXLES TO 32,000 LBS AND GROSS VEHICLE WEIGHT OF 73,280 LBS AUTHORIZED FOR BOTH STANDARD AND LIGHT WEIGHT TRACTOR-TRAILERS WHEN ROUTING IS THROUGH STATES PERMITTING 34 000 LBS ON TANDEM AXLES AND GROSS VEHICLE WEIGHT OF 80,000 LBS

NOTE: WHEN SHIPPING 12 OR LESS UNIT LOADS, BOTH STANDARD AND LIGHT WEIGHT TRACTOR-TRAILERS MAY BE USED IN ALL STATES WITHOUT EXCEEDING THE WEIGHT LAWS

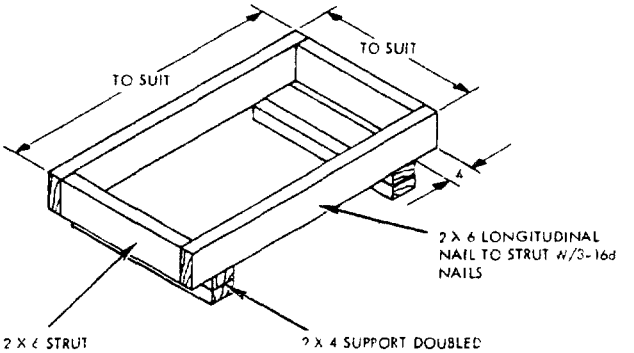
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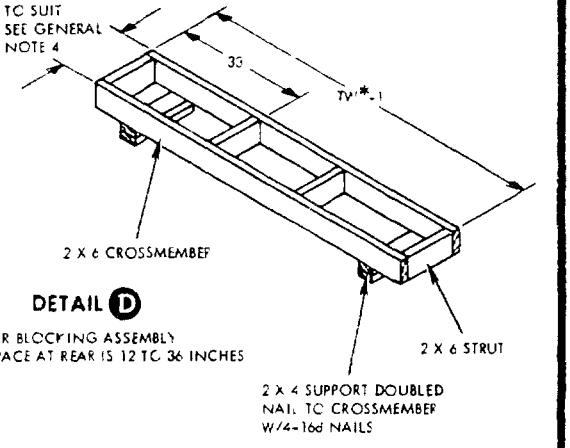
DETAIL A
FRONT BULKHEAD
(1 REQUIRED)



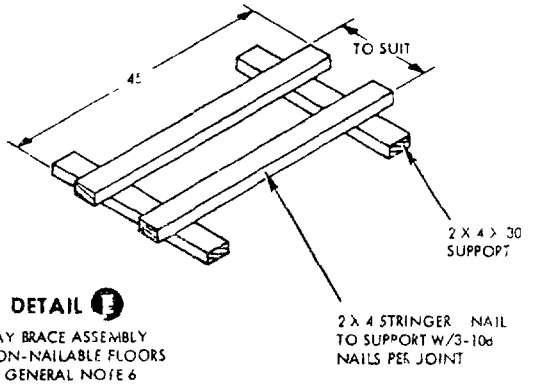
DETAIL B
REAR BLOCKING ASSEMBLY
(1 REQUIRED)
(USED WHEN SPACE AT REAR IS LESS THAN 12 INCHES)



DETAIL C
SIDE BLOCKING ASSEMBLY
FOR NON-NAILABLE FLOORS
SEE GENERAL NOTE 5



DETAIL D
REAR BLOCKING ASSEMBLY
(USED WHEN SPACE AT REAR IS 12 TO 36 INCHES)



DETAIL E
SWAY BRACE ASSEMBLY
FOR NON-NAILABLE FLOORS
SEE GENERAL NOTE 6

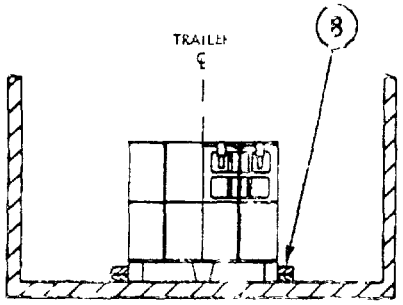
TABLE IV

DOT HAZARD CLASSIFICATION

DODIC / NALC NC	CARTRIDGE TYPE	CARTRIDGE MP-MOD	HAZARD CLASSIFICATION DOT CLASS
AW01	TP	105-0	C
A80c	API	107-0	A
A80F	HEI	106-0&1	A
A811	TP	105-0	C
A812	APT	108-0	C
A871	HEI	106-2	A
A872	API	107-1	A
A873	APT	108-1	C
A874	TP	105-1	C

* TW = TRAILER WIDTH

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SECTION F F

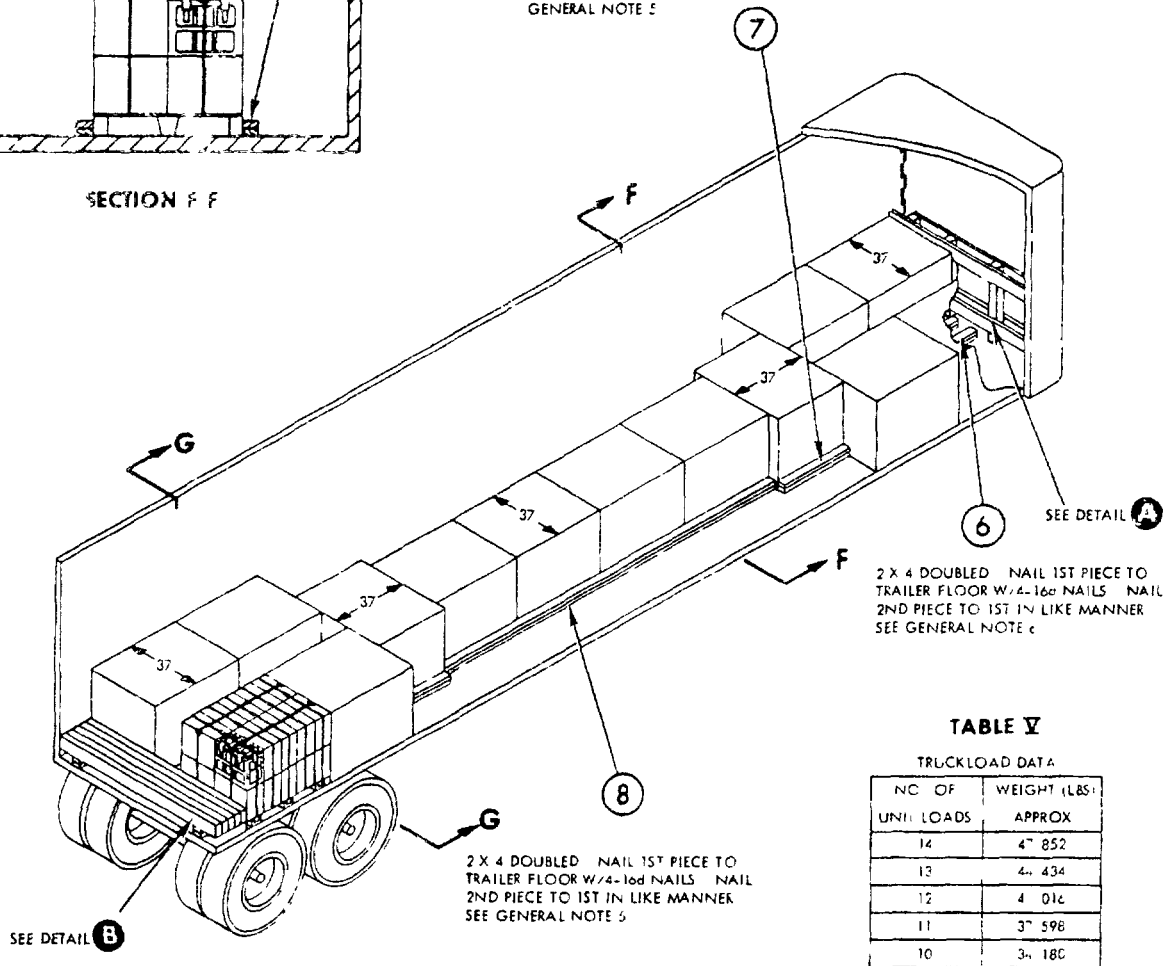
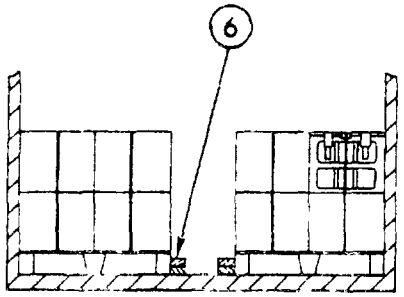


TABLE V

TRUCKLOAD DATA

NO OF UNIT LOADS	WEIGHT (LBS) APPROX
14	47 852
13	44 434
12	41 016
11	37 598
10	34 180
9	30 762



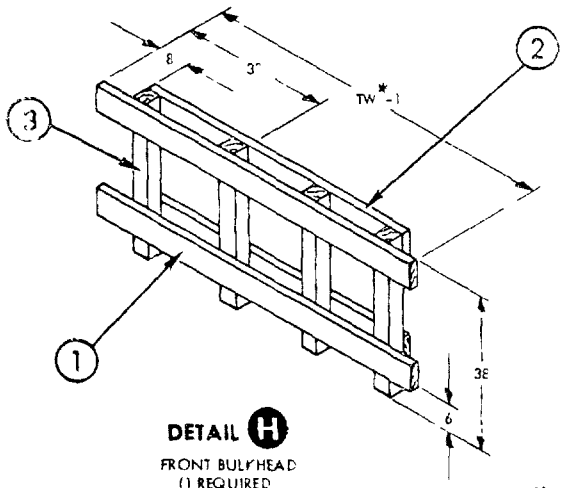
SECTION G G

* TW - TRAILER WIDTH

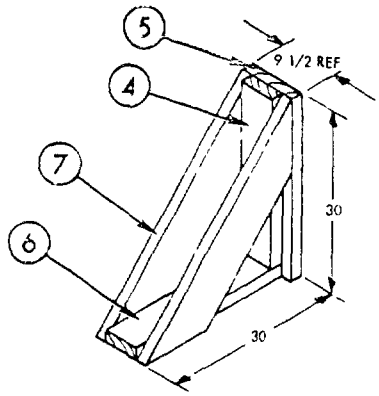
PIECE NO	DESCRIPTION	SIZE	NO OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
8	SLEEPER	2 X 4 X 47	16	SEE FIELD NOTE		
7	SLEEPER	2 X 4 X 37	8	SEE FIELD NOTE		
6	SLEEPER	2 X 4 X 47	16	SEE FIELD NOTE		
5	SUPPORT	2 X 4 TO SUIT	4	4	3	10d
				F	C	10d
4	SOLID FILL	THICKNESS TO SUIT X 6 X TW*-1	AS REQD	LAMINATE	16/PIECE	TO SUIT
3	VERTICAL	4 X 4 X 36	4	SEE 1 & 2	–	–
2	FWD CROSSMEMBER	2 X 6 TO SUIT	2	3	3/JOINT	12d
1	AFT CROSSMEMBER	2 X 6 X TW*-1	2	3	3/JOINT	12d

LIST OF MATERIALS & NAILING DATA

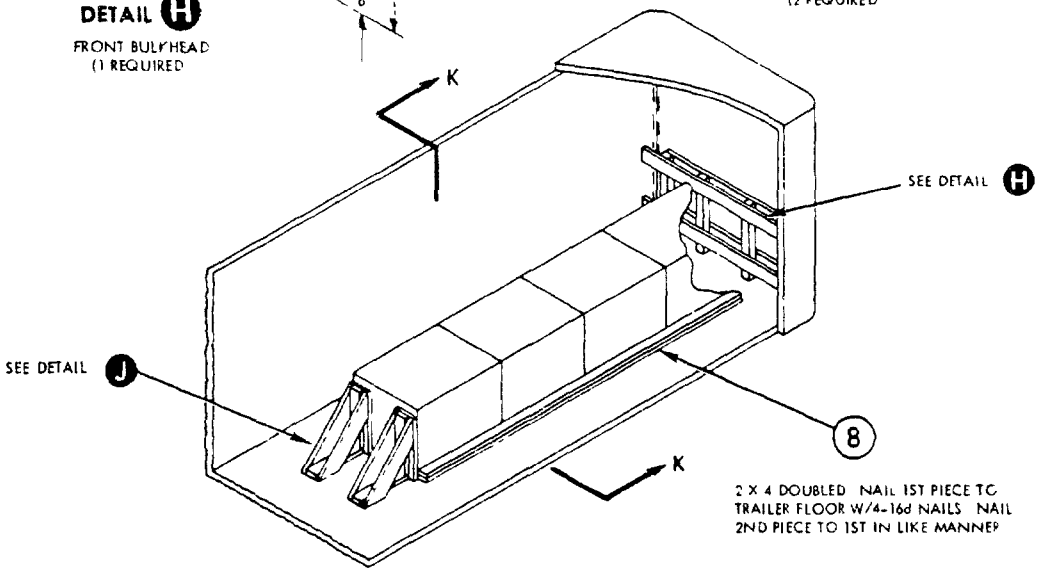
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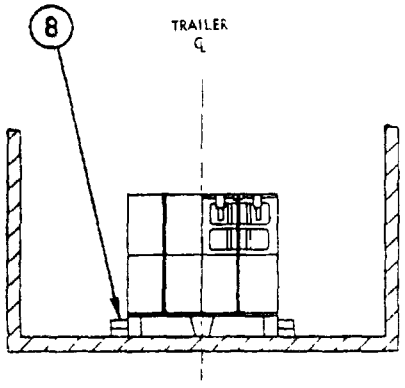
DETAIL H
FRONT BULKHEAD
(1 REQUIRED)



DETAIL J
KNEE BRACE
(2 REQUIRED)



2 X 4 DOUBLED NAIL 1ST PIECE TO TRAILER FLOOR W/4-16d NAILS NAIL 2ND PIECE TO 1ST IN LIKE MANNER



SECTION K-K

* TW = TRAILER WIDTH

PIECE NO	DESCRIPTION	SIZE	NO OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
8	SLEEPER	2 X 4 X 4"	16	SEE FIELD NOTE		
7	BRACE, DIAGONAL	2 X 6 TO SUIT	4	4 & 6	4/JOINT	16d
6	BRACE CLEAT	2 X 6 X 28 1/2	2	TRAILER FLOOR	6	16d
5	BRACE BACKUP	2 X 10 X 30	2	6	3	16d
4	BRACE UPRIGHT	2 X 6 X 28 1/2	2	5	4	8d
3	VERTICAL	4 X 4 X 38	4	SEE 1 & 2	–	–
2	FWD CROSSMEMBER	2 X 6 TO SUIT	2	3	3/JOINT	12d
1	AFT CROSSMEMBER	2 X 6 X TW*-1	2	3	3/JOINT	12d

LIST OF MATERIALS & NAILING DATA

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